



BINURA AMARASINGHE | Data Analyst

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SUMMARY

A responsible, results-oriented, and collaborative professional with a strong foundation in data science, statistical modeling, and business analytics. Skilled in building data-driven solutions to solve complex problems, optimize performance, and support strategic decision-making. Experienced in developing predictive models, conducting market segmentation, and performing simulations using advanced machine learning techniques and BI tools. Committed to continuous learning and applying analytical thinking to deliver actionable insights that drive organizational success.

EXPERIENCE

Intern Data Analyst - Mobitel Pvt Ltd ----- 2024 Nov – 2025 Nov ----

Time Series Forecasting & Revenue Modeling

- Conducted **Time Series Analysis** to identify macroeconomic and micro-level factors (e.g., inflation, exchange rates, media spend) influencing district-level revenue.
- Built and evaluated predictive models using **ARDL, Random Forest, and XGBoost** to capture short- and long-term trends.
- Performed **data normalization, PCA, and multicollinearity checks** to improve model accuracy and interpretability.

Customer Segmentation & Clustering

- Applied **K-Means, Hierarchical Clustering, and K-Modes** to segment regions to identify the districts for each cluster and what are the important features for each cluster.
- Created actionable insights for market segmentation and strategic decision-making.

Monte Carlo Simulation for Hiring & Package Profitability

- Developed **Monte Carlo simulations** to determine optimal combinations of Managers, Senior Executives, and Executives under budget constraints to maximize profit.
- Modeled customer behavior and package transitions to evaluate the financial feasibility of launching a new telecom plan based on churn rates and revenue projections.

Web Scraping & Network Quality Analysis

- Built an automated data pipeline using **Selenium** to scrape hotel listings and reviews from Google Maps and other platforms.
- Filtered and analyzed reviews to extract telecom-related complaints using custom keywords logic to assess customer sentiment on internet and Wi-Fi services.
- Conducted a Booking.com competitor analysis for Amaya Hills Kandy, **scraping recent reviews, prices, ratings, and coordinates** to identify nearby (≤ 5 km) competitors and perform **thematic sentiment analysis** for competitive insights.

Market Potential & Reseller Performance Analysis

- Conducted **RFM (Recency, Frequency, Monetary) analysis** to evaluate and segment over 300+ reseller shops.

- Used **Entropy Weighting Method** to compute a Market Potential Index across districts and guide strategic investment decisions

Tools & Technologies: Python (Pandas, NumPy, Scikit-learn, Matplotlib, Streamlit), Tableau, Power BI, Excel, Selenium, Time Series Models, Machine Learning (XGBoost), PCA, Clustering Techniques.

RESEARCH

Voice-Based AI Voice Assistant for Sri Lanka Call Center Automation ----- 2025 June – 2026 June
Final Year Research Project | Python , RAG, WavLM, LoRA, FAISS

- Built an end-to-end English voice pipeline for a multilingual (Sinhala/English) AI call center assistant, automating customer interactions with emotion-aware responses.
- **End to End English call handling:** Speech recognition, text and audio emotion classification fused using a confidence-adaptive weighting strategy, and full response generation pipeline.
- Implemented a RAG-based response pipeline with intent classification feeding into an LLM for context-aware answer generation.
- Developed an emotion-adaptive TTS system that dynamically adjusts voice characteristics based on detected customer emotion for natural, empathetic interactions.
- Automated CSAT scoring by analyzing conversational patterns throughout each interaction.

PUBLICATIONS

"Multilingual AI Call Center Bot with Sentiment-Aware Dialogue" Accepted for Presentation and Publication — *IEEE ICECET Conference*

PROJECTS

Olympic Data Warehouse Solution | SQL Server, SSIS, SSAS, Power BI

- Designed a snowflake schema with a central fact table and multiple dimensions.
- Built ETL pipelines using SQL Server Integration Services (SSIS) to clean, transform, and load structured/semi-structured data from Excel, CSV, and TXT sources
- Created a multidimensional OLAP cube with SQL Server Analysis Services (SSAS) and defined KPIs for medal performance tracking
- Performed OLAP operations (Slice, Dice, Roll-up, Drill-down, Pivot) using Excel.
- Designed interactive Power BI dashboards with slicers, drill-through filters.

Big Mart Sales Prediction System| Python, Jupyter

- Developed a prediction system to forecast sales for products using python, Jupyter, streamlit
- https://github.com/tharusha01Bandara/Sales_Prediction.git

Recommendation System for E-Commerce | Python, Jupyter

- Developed a hybrid recommendation system for an e-commerce platform using Content-Based and Collaborative Filtering to provide personalized product suggestions.
- https://github.com/thej-k/Personalized_E-commerce_Recommendation_System

EDUCATION

Sri Lanka Institute of Information and Technology (SLIIT) ----- 2022 - 2026
Malabe, Sri Lanka

B.Sc. (Hons) in Information and Technology Specializing in Data Science | CGPA - 3.14

Subjects: Machine Learning, Massive Data & Cloud Computing, Information Retrieval and Web Analytics, Data warehousing and Business Intelligence, Statistical Modelling, Data Structures and Algorithms, Database Management Systems, Internet of Things

University of Kelaniya. ----- 2021-2024

Bachelor of Science (B.Sc.) in Physical Science | CGPA – 2.6

Subjects: Computational mathematics, Mathematical modeling, Numerical methods, Linear algebra, Discrete mathematics, Matrix algebra, Theory of calculus, Mathematics for finance, Differential Equations, Mathematical Methods, Complex Variables, Theory of calculus, Matrix Algebra

University of Colombo School of Computing ----- 2022

Successfully completed Diploma in Bachelor of Information Technology (BIT)

Veyangoda Bandaranayake Central College, Veyangoda. ----- 2019

G.C.E Advanced Level Examination (Physical Science Stream) | Z-Core: 1.3140

Subjects: Combined Mathematics - B, Physics - B, Chemistry – C

TECHNICAL SKILLS

- Programming Languages: Java | Kotlin | Python | R | C | C++
- AI / ML : RAG | LoRA | FAISS | PCA | Clustering | Whisper
- Tools and Platforms: Git | GitHub | Postman
- Version controlling: GitHub
- Database Handling: MySQL | PhpMyAdmin | Firebase | MongoDB
- Web Development: HTML | JS | React JS | Node JS | PHP
- Designing Tools: Draw.io | Figma
- Visualizing Tools: Power BI | Tableau | Things Board

LICENSES AND CERTIFICATIONS

- **SQL for Data Science** - Coursera
- **AI-102: Azure AI Engineer** - Microsoft
- **Fundamentals of Visualization with Tableau** - Coursera
- **AWS Academy Graduate** - AWS Academy Cloud Foundations
- **Successfully completed Certificate course** in English at British council, Sri Lanka

SOFT SKILLS

Analytical and problem-Solving | Research and development | Report writing and presenting |
Hardworking | Teamwork | Decision making | Good communication | Curiosity and passion for learning

EXTRA-CURRICULAR ACTIVITIES

- Active member of mathematics society at the University of Kelaniya.
- Active member of Leo club at University of Kelaniya.

REFEREES

Mr.Sasith Senarath Pathirana

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